EXPLORATORY DATA ANALYSIS

Chapter 20

Sampling, Missingness Patterns, and Data Collection Bias

Understanding who and what the observed data represents — and what it leaves out

Course tools: Python, NumPy, Pandas, Matplotlib's object-oriented API, and Seaborn.

This chapter examines how data is generated, sampled, recorded, and omitted. It focuses on the limits of evidence in an observed dataset, not on a full survey-sampling or causal-inference course.

Learning objectives

• Define a target population, sampling frame, sample, observational unit, and coverage.

• Distinguish probability sampling from non-probability selection and understand inclusion probabilities.

• Recognize coverage, nonresponse, survivorship, selection, and measurement biases.

• Explain MCAR, MAR, and MNAR and identify what can and cannot be inferred from observed data.

• Analyze missingness by variable, row, subgroup, time, and operational source.

• Recognize how filtering, joins, logging rules, and platform behavior shape a dataset.

• Use sensitivity analysis and transparent reporting to communicate limitations.

• Build a reproducible sampling and missingness audit.

> **Central principle:** a dataset is not a neutral window onto reality. It is the result of a collection process. Understanding that process is part of understanding the data.

1. The data-generation and collection chain

Before asking what a dataset says, ask how its records came to exist. A simplified chain is: target phenomenon → eligible units → sampling or selection → participation or capture → measurement → storage and processing → analytical dataset. Bias can enter at every stage.

Stage	Question to ask	Possible failure
Target population	Who do we want to understand?	Population is vague or mismatched to the question
Frame / access	Who could be observed?	Some eligible units are absent from the frame
Selection	Who enters the sample?	Selection depends on behavior or outcome
Response / capture	Who provides usable data?	Nonresponse or logging failure
Measurement	How is a value recorded?	Instrument, wording, proxy, or unit error
Processing	How are records transformed?	Deduplication, joins, filters, or parsing alter coverage
Analysis	Which records are retained?	Complete-case filters or exclusions change the target

A technically clean table can still represent a biased slice of the target population. Data quality and representativeness are related but distinct.

2. Review of earlier chapters (Chapters 1–18)

This review excludes Chapter 19, which is reviewed separately below.

• **Ch. 1 — EDA:** begin with context and questions; observations are not explanations.

• **Ch. 2 — Dataset understanding:** define the observational unit, grain, variable meanings, and data-generating process.

• **Ch. 3 — Data quality:** audit completeness, validity, consistency, uniqueness, accuracy, and timeliness.

• **Ch. 4 — Numerical variables:** use appropriate summaries for scale, spread, and distribution.

• **Ch. 5 — Categorical variables:** inspect frequencies, rare levels, cardinality, and denominators.

• **Ch. 6 — Date/time:** check parsing, time zones, temporal coverage, and event intervals.

• **Ch. 7 — Descriptive statistics:** choose summaries by distribution and purpose.

• **Ch. 8 — Distributions:** inspect tails, gaps, empirical distributions, and density displays.

• **Ch. 9 — Distribution shape:** use skewness, kurtosis, and quantiles with interpretation caveats.

• **Ch. 10 — Distribution visualization:** combine histograms, ECDFs, boxplots, and related displays.

• **Ch. 11 — Relationships:** inspect form, strength, nonlinearity, overplotting, and confounding.

• **Ch. 12 — Multivariate visualization:** use encodings, facets, and heatmaps responsibly.

• **Ch. 13 — Association:** select measures by variable type; association does not establish causation.

• **Ch. 14 — Covariance and dependence:** distinguish zero correlation from independence and inspect conditional structure.

• **Ch. 15 — Grouped analysis:** compare groups with explicit denominators and aggregation choices.

• **Ch. 16 — Outlier detection:** a flag is a candidate for investigation, not an automatic deletion instruction.

• **Ch. 17 — Unusual observations:** investigate source records, peer groups, influence, and evidence.

• **Ch. 18 — High-dimensional EDA:** organize variables, inspect missingness patterns, and avoid over-reading thousands of relationships.

3. Review of Chapter 19 — Time Series EDA

Chapter 19 covered temporal grain, event versus ingestion time, timezone handling, ordering, cadence, missing periods, calendar effects, trend, seasonality, cycles, resampling, rolling summaries, lags, autocorrelation, structural breaks, and panel data.

> **Expected reasoning for the Chapter 19 assignment/project:** a sound analysis defines timestamp semantics and cadence, sorts correctly, distinguishes absent periods from true zero-event periods, justifies aggregation, and interprets unusual periods using operational context. If no assignment was submitted, this is a reasoning guide rather than a grading result.

This chapter extends the same logic: a time series can be perfectly parsed and plotted yet still be biased if outages, reporting delays, or changes in who is observed are ignored.

4. Population, sample, and sampling frame

4.1 Essential definitions

Term	Definition	Example
Target population	The full group the question concerns	All active customers during a stated year
Eligible population	Units meeting explicit inclusion criteria	Customers with an account open on a specified date
Sampling frame	Operational list or mechanism from which units can be selected	Customer registry available to the analyst
Sample	Units actually selected or observed	The 8,000 records extracted
Respondents / captured units	Selected units with usable measurements	Those who answered or generated a record
Analytical sample	Units retained after analysis rules	Rows remaining after a date and completeness filter

These groups need not be identical. The target population may be broader than the frame; the selected sample may be broader than the respondent set; and the analytical sample may be narrower still.

4.2 Define the target before inspecting results

A statement such as “customers” is incomplete. Specify the entity, geography, eligibility, and time window. For example: “people with at least one completed purchase from this platform in Bangladesh between January and December 2025.” Different definitions lead to different populations and conclusions.

4.3 Frame coverage

Coverage error occurs when the frame omits eligible units or includes ineligible ones. An online survey distributed only through an app cannot directly represent people who never use the app, unless the target population is explicitly app users.

5. Probability and non-probability selection

In probability sampling, each unit in the target frame has a known, non-zero selection probability under the design. In non-probability selection, inclusion probabilities are unknown or not governed by a known random design.

Approach	How units enter	Main EDA concern
Simple random sample	Random selection from a frame	Frame completeness and implementation

Approach	How units enter	Main EDA concern
Stratified sample	Random selection within defined strata	Strata definitions and allocation
Cluster sample	Groups are sampled, then units observed	Within-cluster similarity and design effects
Systematic sample	Every kth unit after a start	Ordering or periodicity in the list
Convenience sample	Easy-to-reach units	Access and self-selection
Voluntary response	People choose to participate	Motivation may relate to the topic
Platform / event log	Units that trigger recorded events	Only observable activity is represented

Random sampling does not repair a defective frame, poor measurement, or nonresponse. Conversely, a non-probability dataset can still be useful for describing its observed records; the limitation is how far conclusions can be generalized.

6. Inclusion probabilities and weights — the basic idea

Let π_i be the probability that unit i is included in a probability sample. Units with smaller inclusion probabilities represent more units in the population under a basic design-based perspective. A base weight is often $w_i = 1/\pi_i$. Weighting can help align estimates with the sampling design, but it does not automatically correct coverage error, nonresponse, measurement error, or unknown selection mechanisms.

For this course, the goal is to recognize weights, inspect how they were constructed, and avoid treating a weighted statistic as self-validating. Detailed survey estimation, calibration, and variance estimation are beyond this chapter.

```
# Inspect a supplied weight column; do not invent weights.
weight_summary = df["sample_weight"].describe(
    percentiles=[0.01, 0.25, 0.50, 0.75, 0.99]
)
weight_summary

# Compare an unweighted and weighted mean when the
# weight's meaning and validity are documented.
valid = df[["value", "sample_weight"]].dropna()
weighted_mean = np.average(
    valid["value"], weights=valid["sample_weight"]
)
unweighted_mean = valid["value"].mean()
```

Weights should be non-negative and interpreted according to their documentation. Extreme weights can make estimates unstable. Never infer a valid weighting scheme from the data alone.

7. Major forms of data collection bias

7.1 Coverage bias

Some parts of the target population cannot enter the frame or capture system. Example: a digital service's logs exclude people who cannot access the service.

7.2 Nonresponse bias

Selected units do not provide usable data, and response propensity is related to the quantity being studied. Example: people with very poor service experiences may be more likely to complete a complaint survey—or less likely if they have disengaged.

7.3 Self-selection / voluntary response bias

Participation is chosen by the individual. People with strong opinions, high engagement, or particular experiences may be overrepresented.

7.4 Survivorship bias

Only units that remain observable or successful are included. Studying only businesses still operating omits those that failed; studying only active users omits those who churned.

7.5 Measurement and response bias

The measurement process systematically differs from the true quantity. Leading survey wording, recall error, social desirability, sensor calibration, proxy variables, and inconsistent units can all distort observations.

7.6 Selection on a variable

Filtering or conditioning on a variable influenced by multiple other variables can create or distort associations. For example, studying only people who were approved for a loan may induce relationships among factors that influenced approval.

7.7 Time-window and availability bias

A dataset may overrepresent recent, active, or easily recorded periods. Delayed records, outages, backfills, and changing retention policies can alter the apparent history.

8. Missing data: what is actually missing?

Missingness can mean a value was not asked, not applicable, refused, not measured, not captured, lost in processing, or not yet available. These states should not be collapsed into one undifferentiated null when their meanings differ.

Missingness source	Example	Possible interpretation
Not applicable	Spouse income for an unmarried respondent	Structural absence
Not asked / skip logic	Follow-up shown only after a prior answer	Designed missingness
Refusal / unknown	Respondent declines to state income	Potentially informative
Measurement failure	Sensor did not return a reading	Operational failure
Not captured	Event occurred outside logging coverage	Coverage problem
Processing loss	Join or parse operation drops values	Pipeline defect
Not yet observed	Outcome window has not elapsed	Right-censoring / incomplete follow-up

A missing-value codebook is often more informative than a single missing percentage. Preserve distinct states where possible.

9. MCAR, MAR, and MNAR

These labels describe assumptions about the missingness mechanism, not the percentage of missing values.

Mechanism	Informal meaning	Example
MCAR	Missingness is unrelated to observed or unobserved data	A random storage failure drops a few records
MAR	After conditioning on observed variables, missingness does not depend on the missing value itself	Income is more often missing for a known age group, and age is ob
MNAR	Missingness still depends on the unobserved value, even after observed variables are considered	People with very high income are less likely to disclose income for that reaso

Formally, let R indicate whether Y is observed and let X be observed information. MCAR implies R is independent of both observed and unobserved data. MAR allows R to depend on observed data X but not on the missing value after conditioning on X. MNAR allows dependence on the unobserved value.

> **Important:** MAR and MNAR generally cannot be distinguished from the observed data alone without additional assumptions, external information, follow-up, or sensitivity analysis. A missingness pattern is evidence to investigate, not proof of a mechanism.

10. Quantifying and visualizing missingness

10.1 Column-level missingness

```
missing_by_column = (
    df.isna().sum().rename("missing_n").to_frame()
)
missing_by_column["observed_n"] = len(df) - (
    missing_by_column["missing_n"]
)
missing_by_column["missing_rate"] = (
    missing_by_column["missing_n"] / len(df)
)
missing_by_column = missing_by_column.sort_values(
    "missing_rate", ascending=False
)
```

10.2 Row-level missingness

```
row_missing_n = df.isna().sum(axis=1)
row_missing_rate = df.isna().mean(axis=1)
```

```
row_missing_summary = row_missing_n.value_counts().sort_index()
```

10.3 Missingness by subgroup

Compare missingness rates across groups that are relevant to the collection process: source system, region, survey mode, product, time period, or demographic categories where appropriate and ethically permissible.

```
group_missing = (
    df.assign(income_missing=df["income"].isna())
      .groupby("survey_mode", dropna=False)
      ["income_missing"]
      .agg(["size", "mean"])
      .rename(columns={"size": "n", "mean": "missing_rate"})
)
```

10.4 Missingness over time

```
df["month"] = pd.to_datetime(
    df["timestamp"], errors="coerce"
).dt.to_period("M")

monthly_missing = (
    df.groupby("month")["income"]
      .apply(lambda s: s.isna().mean())
)
```

A sudden increase in missingness may indicate a form change, source outage, policy change, or a shift in the population being observed.

11. Co-missingness and missingness indicators

When several fields are missing together, they may share a questionnaire branch, system, device, or collection process. A missingness indicator can be useful for describing these patterns, but it should not be mistaken for the missing value itself.

```
missing_flags = df.isna().astype("int8")
missing_cooccurrence = missing_flags.T @ missing_flags

# Pairwise overlap of missingness indicators:
missing_corr = missing_flags.corr()
```

Co-occurrence counts depend on dataset size and column prevalence. Correlation of binary missingness indicators is also affected by prevalence. Inspect counts, rates, and the operational meaning together.

12. Complete-case analysis and its risks

Complete-case analysis removes every row with a missing value in any selected variable. It is simple, but it changes the analytical sample and may reduce precision or introduce bias.

```
analysis_columns = ["age", "income", "region", "outcome"]
complete = df.dropna(subset=analysis_columns)

retention = len(complete) / len(df) if len(df) else np.nan
print("Rows retained:", len(complete))
print("Retention rate:", retention)
```

If missingness is MCAR, complete-case estimates may remain unbiased for some estimands, though less precise. Under MAR or MNAR, bias can occur depending on the analysis and variables. Even under MCAR, dropping rows can waste information.

Always report how many records were removed, which fields drove removal, and how retained records differ from excluded records on observed characteristics.

13. Selection effects, conditioning, and collider intuition

A selection variable can be influenced by two other variables. Restricting analysis to units with a particular selection value can create an association between those causes, even if they were unrelated in the full population. This is one route to selection bias.

Example: both financial stability and health status may influence whether someone participates in a demanding study. Among participants only, financial stability and health may appear related because both affect selection. The observed association need not reflect the population relationship.

For EDA, document filters and eligibility rules, compare included and excluded records where possible, and be cautious when interpreting associations inside a selected subset. A causal diagram can help formulate the concern, but causal identification is outside this chapter.

14. Joins, duplicates, and pipeline-induced selection

Data engineering operations can alter the population represented by a table.

• **Inner join:** drops records without a match on both sides; unmatchedness may be systematic.

• **Left join:** preserves the left table but may create missing values where the right side has no match.

• **One-to-many join:** can multiply rows and overrepresent entities with more matches.

• **Deduplication:** may remove legitimate repeated events if the key is poorly chosen.

• **Filtering:** date, status, or completeness rules can exclude meaningful segments.

• **Late-arriving data:** recent periods may appear incomplete until backfill finishes.

```
# Audit key coverage before relying on an inner join.
left_keys = set(left["customer_id"].dropna().unique())
right_keys = set(right["customer_id"].dropna().unique())

left_unmatched = left_keys - right_keys
right_unmatched = right_keys - left_keys

joined = left.merge(
    right, on="customer_id", how="left",
    validate="many_to_one", indicator=True
)
join_audit = joined["_merge"].value_counts(dropna=False)
```

Use merge validation when the expected relationship is known. Check row counts and key coverage before and after each join.

15. Comparing observed and excluded records

A practical diagnostic is to compare records retained for analysis with records excluded by a filter or missingness rule. Differences do not prove bias, but they reveal how the analytical sample changes.

```
work = df.copy()
work["complete_for_analysis"] = (
    work[analysis_columns].notna().all(axis=1)
)

comparison = work.groupby(
    "complete_for_analysis"
).agg(
    n=("customer_id", "size"),
    mean_age=("age", "mean"),
    median_age=("age", "median")
)
```

Choose comparison variables that are available for both groups. If the key variable itself is missing, you cannot directly compare its distributions without additional assumptions or information.

16. Sensitivity analysis and responsible conclusions

When the mechanism is uncertain, do not pretend one assumption is known. Instead, ask whether the main descriptive conclusion changes under plausible alternatives.

• Compare the full observed sample with complete cases.

• Report subgroup-specific missingness and retention rates.

• Compare results across reasonable time windows or source systems.

• Check whether a pattern is driven by one region, platform, or high-response group.

• Use externally documented population totals or frame information when available.

• State what cannot be learned from the observed records alone.

Sensitivity analysis does not magically remove bias. It makes dependence on assumptions visible.

17. A reusable sampling and missingness audit

```
def collection_audit(
    df, key=None, group_cols=None, time_col=None
):
    n = len(df)
    report = {
        "rows": n,
        "columns": len(df.columns),
        "missing_by_column": df.isna().mean().sort_values(
            ascending=False
        ),
        "missing_fields_per_row": df.isna().sum(axis=1),
        "complete_row_rate": (
```

```
            df.notna().all(axis=1).mean() if n else np.nan
        )
    }

    if key is not None:
        report["duplicate_key_rows"] = int(
            df.duplicated(key, keep=False).sum()
        )

    if group_cols:
        for col in group_cols:
            report[f"missing_by_{col}"] = (
                df.assign(_missing=df.isna().sum(axis=1))
                  .groupby(col, dropna=False)["_missing"]
                  .agg(["size", "mean"])
            )

    if time_col:
        times = pd.to_datetime(df[time_col], errors="coerce")
        report["invalid_timestamps"] = int(times.isna().sum())
        report["time_min"] = times.min()
        report["time_max"] = times.max()

    return report
```

Extend this audit with source-system counts, eligibility rules, frame coverage, response status, join diagnostics, and documented definitions. The function cannot infer representativeness from a DataFrame.

18. Common mistakes

• Assuming a large dataset is automatically representative.

• Using “random” to describe a convenience or platform sample.

• Confusing the target population with the records available in a table.

• Treating MCAR, MAR, or MNAR as conclusions established by a missingness heatmap.

• Assuming a missing percentage reveals why values are missing.

• Replacing all missing values with zero without checking meaning.

• Dropping incomplete rows without reporting retention and sample changes.

• Ignoring people, events, or periods absent from the collection frame.

• Using an inner join without checking which records disappear.

• Treating a weighted statistic as correct without understanding the weights.

• Making population-wide claims from a selected or survivorship-filtered dataset.

19. Summary

• Every dataset is produced by a chain of target definition, coverage, selection, response, measurement, processing, and analysis.

• Define the target population, frame, sample, respondents, and analytical sample separately.

• Distinguish probability sampling from non-probability selection.

• Recognize coverage, nonresponse, self-selection, survivorship, measurement, and pipeline bias.

• Interpret missingness in context; MCAR, MAR, and MNAR are mechanism assumptions.

• Measure missingness by column, row, subgroup, source, and time.

• Audit joins, filters, duplicate handling, and late-arriving data.

• Use sensitivity analysis and state limitations instead of overstating generalizability.

20. Assignment

Part A — Concepts

• Define a target population, sampling frame, sample, respondent set, and analytical sample for a dataset you know.

• Give one realistic example each of coverage bias, nonresponse bias, survivorship bias, and measurement bias.

• Explain MCAR, MAR, and MNAR using one example from a survey or operational dataset.

• Why can a large convenience sample still be unrepresentative?

• Explain how an inner join can create selection in an analytical dataset.

Part B — Collection-process map

Choose a dataset and write a short collection-process description: target population, eligibility, frame or capture mechanism, selection, measurement, processing, and analysis filters. Mark which stages are known, uncertain, or undocumented.

Part C — Missingness audit

• Report missing counts, rates, and denominators by column.

• Summarize missing fields per row.

• Compare missingness across at least two meaningful subgroups and, where possible, across time.

• Inspect co-missingness for a selected group of variables.

• Compare retained and excluded records using variables observed in both groups.

Part D — Bias assessment

• Identify at least three plausible biases in the collection process.

• For each, explain which units or values may be over- or under-represented.

• State which population-level conclusions are defensible and which are not.

• Propose additional data or process documentation that would reduce uncertainty.

Part E — Reflection

Explain why the absence of evidence in your dataset is not always evidence that an event, person, or value did not exist.

21. Coding project — Data Collection and Missingness Audit

Extend your cumulative EDA master notebook. The project should combine technical diagnostics with a clear account of how the dataset came to exist.

• Document the target population, observational unit, eligibility criteria, sampling frame or capture mechanism, and time window.

• Map the path from selection and response through measurement, storage, joins, and analytical filters.

• Build column-, row-, subgroup-, and time-based missingness summaries.

• Preserve distinct missing states when the source distinguishes refusal, not applicable, not measured, and not captured.

• Audit key uniqueness, join coverage, row-count changes, and filters.

• Compare included and excluded records on observed variables.

• Identify plausible coverage, nonresponse, survivorship, measurement, and pipeline biases.

• Explain whether supplied weights exist, what they mean, and what limitations remain.

• Perform at least two sensitivity comparisons.

• Write a professional report separating observed facts, plausible mechanisms, assumptions, and unknowns.

Acceptance criteria: every denominator is explicit; sample definitions are reproducible; missingness is not automatically imputed or erased; joins and filters are audited; bias claims are framed as evidence-based possibilities; and population claims match the available sampling information.

Suggested deliverables: updated master notebook, collection-process map, missingness audit tables, join diagnostics, selected figures, and a concise limitations section suitable for a client-facing EDA report.

22. Preview of Next Chapter

Chapter 21 — Automated EDA and Data Profiling

We will examine tools that automate parts of exploratory analysis: profiling reports, schema and quality summaries, distribution and association scans, warnings, and reproducible report generation. We will learn what automated EDA can accelerate, what it can miss, and how to review its output critically.

Boundary: automated tools will support—not replace—question-driven analysis, domain understanding, and professional interpretation.